# Package 'iBHMM'

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**Title** iBHMM: Bayesian Supervised Biclustering via Hidden Markov Models

**Version** 0.1.1

**Description** Implements a Bayesian supervised biclustering framework
using Hidden Markov Models for multi-modality data integration.
The model performs joint clustering and outcome modeling via
Gibbs sampling.

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## R topics documented:


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`Cindex` *Concordance Index (C-index) for Censored Outcomes*

---

### Description

Computes the concordance index using predicted times, observed times, and event indicators.

### Usage

```
Cindex(Tpredict, Ttest, Status)
```

### Arguments

| | |
|---|---|
| Tpredict | Numeric vector of predicted times. |
| Ttest | Numeric vector of observed times. |
| Status | Integer/numeric vector of event indicators (1 = event observed, 0 = censored). |

### Value

Numeric scalar concordance index.

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## EvaluateTest *Evaluate Test-Set Predictions*

---

### Description

Computes an outcome-specific performance metric using posterior predictive samples returned by iBiclust.

### Usage

```
EvaluateTest(result, TestY, OutcType, threshold = 0, return_all = FALSE)
```

### Arguments

| | |
|---|---|
| result | A fitted model object returned by iBiclust containing YpredTestSample (posterior predictive samples for test outcomes). |
| TestY | Test-set response variable. For continuous or binary outcomes, this should be a numeric vector. For censored outcomes, this must be an $n \times 2$ matrix of the form cbind(time, censor), where the second column is a censoring indicator (1 = censored, 0 = event observed). |
| OutcType | Outcome type: one of "continuous", "binary", or "censored". |
| threshold | Numeric; classification threshold for binary outcomes applied to the posterior predictive mean. Default is 0. |
| return_all | Logical; if TRUE, returns a list including intermediate predictions. If FALSE, returns only the metric(s). |

### Value

For OutcType = "continuous": a scalar mean square error MSE. For OutcType = "binary": a scalar misclassification error MCE. For OutcType = "censored": a scalar concordance index Cindex.

### Examples

```
## Not run:
## Assume 'result' is a fitted iBiclust model with test predictions
## and 'datTest$Y' is the corresponding test response.

metric <- iEvaluateTest(
  result = result,
  TestY = datTest$Y,
  OutcType = "continuous"
```

```
)

metric

## End(Not run)
```

---

`iBiclust` *Bayesian Supervised Biclustering via Hidden Markov Models for Multi-Modal Data*

---

## Description

Fits a Bayesian model-based supervised biclustering model for multi-modality datasets using an Gibb sampling algorithm. Each modality/platform is provided as a matrix with samples in rows and features in columns; internally, each matrix is optionally standardized (centered/scaled) and transposed before being passed to the native routine.

## Usage

```
iBiclust(
  Enable_HMM = TRUE,
  dataList = dataList,
  OutcType = "continuous",
  Youtcome = Youtcome,
  Cov = NULL,
  sample = 2000,
  burnin = 1000,
  K = 2,
  normalized = TRUE,
  seed = 1,
  hyperSigma = c(1, 1),
  alphaP = rep(1, K),
  delt = rep(1, 3),
  TruncVal = 0.1,
  dataListTest = NULL,
  CovTest = NULL,
  return_samples = FALSE
)
```

## Arguments

| | |
|---|---|
| `Enable_HMM` | Logical; if `TRUE` (default), the model incorporates a Hidden Markov Model (HMM). If `FALSE`, the model does not incorporate the HMM structure. |
| `dataList` | A list of modality matrices. Each `dataList[[t]]` is an $n \times p_t$ matrix with samples in rows and features in columns. |
| `OutcType` | Outcome type: one of `"continuous"`, `"binary"`, or `"censored"`. |
| `Youtcome` | Outcome vector/matrix. For censored outcomes, this must be an $n \times 2$ matrix `cbind(time, censor)`, where the second column represents the *censoring indicator* (1 = censored, 0 = event observed). |
| `Cov` | Covariate matrix of dimension $n \times p_c$. |

| | |
|---|---|
| sample | Number of MCMC samples to retain. |
| burnin | Number of burn-in iterations. |
| K | Number of sample clusters used in the model. |
| normalized | Logical; if TRUE, each modality is centered and scaled feature-wise. |
| seed | Integer random seed passed to the native routine. |
| hyperSigma | Numeric vector of length 2. The prior distribution on the state-specific variances $\sigma^2_{dk1}$ and $\sigma^2_{dk2}$ for $\mu_{d,j,k,l}$. |
| alphaP | Numeric vector of length K specifying the Dirichlet prior concentration parameters for the cluster probability vector. These parameters control the prior probability that a sample belongs to each cluster $k$. |
| delt | Numeric vector of length 3 giving the Dirichlet prior concentration parameters $(\delta_1, \delta_2, \delta_3)$ for each row of the transition matrix $\xi$. |
| TruncVal | Positive numeric truncation threshold $t$ used in the Gibbs sampler. |
| dataListTest | Optional test set list with same structure as dataList. |
| CovTest | Optional test covariate matrix of dimension $n_{test} \times p_c$. |
| return_samples | Logical; additionally return ZSample, probZSample, MuSample, and probKappaSample when TRUE (default FALSE). |

## Details

**Data format:**

- dataList must be a list of length M (number of modalities/platforms).
- Each element dataList[[t]] is an $n \times p_t$ matrix: $n$ samples (rows) and $p_t$ features (columns).
- If normalized = TRUE, each modality in dataList is centered and scaled feature-wise using its own mean and standard deviation. If dataListTest is provided, the test data are standardized using the corresponding training means and standard deviations (not re-estimated from the test data).

**Ordering requirement (HMM):** If Enable_HMM = TRUE (Hidden Markov Model), the features within each modality in dataList are assumed to be pre-ordered according to their natural or genomic sequence. The function does not reorder features internally.

**Outcome types:**

- "continuous": OutcomeType = 1
- "binary": OutcomeType = 2
- "censored": OutcomeType = 3, where Youtcome must be an $n \times 2$ matrix (e.g., value and censoring indicator).

**Test data:** If dataListTest is provided, it must have the same list structure as dataList. If normalized = TRUE, test data are standardized using the training means/SDs.

## Value

A list with posterior summaries, coefficient draws, test labels and predictions, and log-posterior values. Only ZSample, probZSample, MuSample, and probKappaSample require return_samples = TRUE. Label switching is checked automatically. If label switching occurs, cluster labels are aligned using iterative ECR:

**ZSample, probZSample, MuSample, probKappaSample** Retained labels, cluster probabilities, latent means, and feature-state probabilities when requested.

**DIC** Deviance Information Criterion returned by the sampler.

**probZMean** Matrix of posterior mean cluster membership probabilities ($n \times K$).

**MuMean** Array of posterior means of the latent mean parameters $\mu_{dkl}$ for the over- and under-expressed states ($l = 1, 2$), indexed by modality $d$ and cluster $k$.

**CoefySample** cluster-specific outcome means with dimension $sample \times K$.

**sig2yM** Posterior mean of outcome variance parameter.

**KappaMean** List of arrays of posterior means for biclustering indicators per modality.

**TransMatMean** Array of posterior mean transition matrices per modality.

**Betasample** Matrix of sampled covariate coefficients ($sample \times p_c$).

**ZtestSample** Matrix of sampled test cluster labels ($sample \times n_{test}$) if test data provided.

**YpredTestSample** Matrix of sampled test predictions ($sample \times n_{test}$) if test data provided.

**logpostsample** Vector of log-posterior values of length `sample` + `burnin`.

## Examples

```
## Not run:
## Generate synthetic multi-modality data
K <- 2
p <- c(300, 500)   # smaller for an example
n <- 100
nbrcov <- 4
OutcType <- "continuous"

dat <- iGenDataHMM(
  OrderFeatured = TRUE, nActive = 100, K = K, NbrPlat = 2, OutcType = OutcType,
  MeanY = c(rep(1, floor(K/2)), rep(-1, K - floor(K/2))),
  pc = nbrcov, betacov = 1.0,
  MeanOverExpre = (1:K) / 2, MeanUnderExpre = -(1:K) / 2,
  p = p, seed = 1, n = n,
  SigmaOE = rep(1, K), SigmaUE = rep(1, K)
)

## Generate test data
ntest <- 100
datTest <- iGenDataHMM(
  OrderFeatured = TRUE, nActive = 100, K = K, NbrPlat = 2, OutcType = OutcType,
  MeanY = c(rep(1, floor(K/2)), rep(-1, K - floor(K/2))),
  pc = nbrcov, betacov = 1.0,
  MeanOverExpre = (1:K) / 2, MeanUnderExpre = -(1:K) / 2,
  p = p, seed = 2, n = ntest,
  SigmaOE = rep(1, K), SigmaUE = rep(1, K)
)

## Fit model (keep MCMC small in examples)
result <- iBiclust(
  Enable_HMM = TRUE,
  dataList = dat$X,
  Youtcome = dat$Y,
  OutcType = OutcType,
  Cov = dat$Cov,
```

```
  sample = 200,
  burnin = 100,
  K = K,
  normalized = FALSE,
  seed = 1,
  hyperSigma = c(1, 1),
  alphaP = rep(1, K),
  delt = rep(1, 3),
  TruncVal = 0.1,
  dataListTest = datTest$X,
  CovTest = datTest$Cov
)

str(result)

## End(Not run)
```

---

`iGenDataHMM` *Generate Synthetic Multi-Modal Data for HMM-Based Supervised Biclustering*

---

## Description

Simulates multi-modality feature matrices, covariates, and an outcome under a cluster/state structure used by `iBiclust`. For each modality, features are assigned to one of three states within each cluster: over-expressed (`1`), under-expressed (`2`), or baseline (`3`). Outcomes can be continuous, binary, or censored.

## Usage

```
iGenDataHMM(
  OrderFeatured = FALSE,
  K = K,
  NbrPlat = 2,
  OutcType = "continuous",
  MeanY = MeanY,
  pc = NULL,
  betacov = 1,
  MeanOverExpre = 1:K,
  MeanUnderExpre = -c(1:K),
  p = p,
  seed = 1,
  n = n,
  SigmaOE = rep(1, K),
  SigmaUE = rep(1, K),
  nActive = 100L
)
```

## Arguments

| | |
|---|---|
| `OrderFeatured` | Logical; if `TRUE`, uses a deterministic ordering for feature membership in each modality (the first nActive features over-expressed and the next nActive under-expressed). If `FALSE`, feature membership is randomized. |

| | |
|---|---|
| K | Integer; number of clusters. |
| NbrPlat | Integer; number of modalities/platforms/views. |
| OutcType | Outcome type: one of "continuous", "binary", or "censored". |
| MeanY | Numeric scalar, length-K vector, or NbrPlat by K matrix; cluster-specific mean outcome levels. |
| pc | Integer; number of covariates to generate. Must be a positive integer. |
| betacov | Numeric; magnitude of the covariate coefficients. Coefficients are assigned as +(betacov) for the first half and -(betacov) for the remainder. |
| MeanOverExpre | Numeric scalar, length-K vector, or NbrPlat by K matrix; means for over-expressed features in each cluster. |
| MeanUnderExpre | Numeric scalar, length-K vector, or NbrPlat by K matrix; means for under-expressed features in each cluster. |
| p | Integer vector of length NbrPlat; number of features per modality. |
| seed | Integer; random seed used for simulating X and outcomes. |
| n | Integer; number of samples. |
| SigmaOE | Numeric scalar, length-K vector, or NbrPlat by K matrix; standard deviations for over-expressed features. |
| SigmaUE | Numeric scalar, length-K vector, or NbrPlat by K matrix; standard deviations for under-expressed features. |
| nActive | Positive integer or length-NbrPlat vector; number of features in each active state per modality (default 100). |

## Details

**Cluster assignment:** Samples are assigned to K clusters in approximately equal-sized blocks.

**Feature membership (kappajk):** For each modality $m$, a $p_m \times K$ matrix is generated with entries in {1,2,3}. If OrderFeatured = TRUE, the first nActive features are set to state 1 (over-expressed) and the next nActive features to state 2 (under-expressed). Otherwise, nActive over- and nActive under-expressed features are sampled at random for each modality (requires p[m] >= 2 * nActive[m]).

**Outcome generation:** A latent mean MeanY[Zk[i]] is assigned to each sample and combined with a linear covariate effect. Continuous outcomes add Gaussian noise; binary outcomes threshold a noisy latent score; censored outcomes return an $n \times 2$ matrix cbind(time, censor), where the second column is a censoring indicator (1 = censored, 0 = event observed).

## Value

A list with components:

**Y** Outcome vector of length n for continuous/binary outcomes, or an $n \times 2$ matrix for censored outcomes cbind(time, censor), where the second column is the censoring indicator (1 = censored, 0 = event observed).

**X** List of length NbrPlat; each element is an $n \times p_m$ matrix of simulated features.

**kappajk** List of length NbrPlat; each element is a $p_m \times K$ matrix with entries in {1,2,3} indicating over/under/baseline feature membership for each cluster.

**Zk** Integer vector of length n giving true cluster assignments in 1:K.

**Cov** Numeric matrix of dimension $n \times pc$ containing generated covariates.

## Examples

```
set.seed(1)
K <- 2
p <- c(300, 500)
n <- 100
dat <- iGenDataHMM(
  OrderFeatured = TRUE, nActive = 100, K = K, NbrPlat = 2,
  OutcType = "continuous", MeanY = c(1, -1), pc = 4, betacov = 1.0,
  MeanOverExpre = (1:K) / 2, MeanUnderExpre = -(1:K) / 2,
  p = p, seed = 1, n = n,
  SigmaOE = rep(1, K), SigmaUE = rep(1, K)
)
str(dat)
```

---

# ShowBicluterMeanTrace *Plot MCMC Traces of Bicluster Means*

---

## Description

Creates trace plots for the overexpressed and underexpressed bicluster means retained by `iBiclust`. The plot shows retained iteration on the horizontal axis and mean parameter value on the vertical axis, with colors identifying clusters and panels identifying modalities and expression states.

## Usage

```
ShowBicluterMeanTrace(result)
```

## Arguments

| | |
|---|---|
| `result` | Results returned by `iBiclust` with `return_samples = TRUE`. Its `MuSample` component must be an array with dimensions retained iterations by modalities by clusters by expression states, where state 1 is overexpressed and state 2 is underexpressed. |

## Value

A list with two components:

**data** A long-format data frame containing the retained iteration `id`, cluster `k`, modality label `d`, expression state `name`, sampled mean `value`, modality/cluster label `group`, and `chain` (set to 1).

**figure** A `ggplot` object. Use `print()` to display it.

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